GUIDE 3 OF 3

Spotting AI Mistakes in Mathematics

Six errors that chat assistants make with confidence, how to catch each one, and a checklist.

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A language model predicts text that looks like a solution. Most of the time that text is right, which is exactly why the wrong cases are dangerous: they look the same. None of the mistakes below comes with a warning. Each example is small enough to check by hand, and you should.

Six typical mistakes

1. A sign error in an inequality

The model writes: Solve $3 - 2x > 7$. Subtract 3: $-2x > 4$. Divide by -2: $x > -2$.

How to catch it: Test a value from the answer. $x = 0$ satisfies $x > -2$, but $3 - 0 = 3$ is not greater than 7.

Correct: dividing by a negative number reverses the inequality, so $x < -2$. Check: $x = -3$ gives $3 + 6 = 9 > 7$.

2. A dropped case

The model writes: Solve $x^2 > 9$. Take square roots: $x > 3$.

How to catch it: Try a negative number. $x = -4$ gives $16 > 9$, yet -4 is not in the answer.

Correct: $\sqrt{x^2} = |x|$, so $|x| > 3$, that is $x < -3$ or $x > 3$.

3. Ignoring the domain

The model writes: Solve $\log x + \log(x - 3) = \log 4$. Combine: $x(x - 3) = 4$, so $x^2 - 3x - 4 = (x - 4)(x + 1) = 0$ and $x = 4$ or $x = -1$.

How to catch it: Write the domain before you start: both logarithms need positive arguments, so $x > 3$. Then substitute each root back in: $\log(-1)$ is not defined.

Correct: $x = 4$ only. Check: $\log 4 + \log 1 = \log 4$. The same trap appears when you square both sides: $\sqrt{x + 2} = x$ gives $x^2 - x - 2 = 0$, with roots 2 and -1, but only $x = 2$ works, because $\sqrt{1} = 1 \neq -1$.

4. A real theorem used without its hypotheses, or an invented one

The model writes: By the Mean Value Theorem, every continuous function on $[0, 1]$ is differentiable at some point of $(0, 1)$.

How to catch it: Look the theorem up and read its hypotheses. The Mean Value Theorem *assumes* that f is differentiable on $(0, 1)$; it cannot prove it.

Correct: the claim is false. Weierstrass constructed a function that is continuous everywhere and differentiable nowhere. The same check applies to references: a model can produce a paper with plausible authors, journal, and year that does not exist. Search the title or the DOI before you cite it.

5. A confident arithmetic slip

The model writes: $1 + 2 + 3 + \cdots + 100 = 5500$.

How to catch it: Use a second method. Pair the first and last terms: $1 + 100$, $2 + 99$, … gives 50 pairs, each summing to 101. Or use a CAS.

Correct: $100 \cdot 101/2 = 5050$. A model may also get the method right and the final multiplication wrong, so redo the last line yourself.

6. A circular proof

The model writes: Prove $a^2 + b^2 \geq 2ab$ for all real a, b. Since $a^2 + b^2 \geq 2ab$, we get $a^2 - 2ab + b^2 \geq 0$, that is $(a - b)^2 \geq 0$, which is true. □

How to catch it: Ask what the first line assumes. Here it assumes the statement to be proved. Deriving a true statement from a claim does not prove the claim: from $1 = 2$ you can derive $0 \cdot 1 = 0 \cdot 2$, which is true.

Correct: start from what you know and end at the claim. For all real a, b, $(a - b)^2 \geq 0$. Expanding gives $a^2 - 2ab + b^2 \geq 0$, so $a^2 + b^2 \geq 2ab$. □

Checklist before you trust an answer

- ☐ I substituted the answer back into the original problem.
- ☐ I tested at least one value inside and one outside the solution set.
- ☐ I wrote down the domain before solving, and every root lies in it.
- ☐ Every time I multiplied or divided an inequality, I checked the sign of the factor. Every time I divided, I checked it could not be zero.
- ☐ If I squared both sides, I checked every root in the original equation.
- ☐ I looked up every named theorem and checked its hypotheses.
- ☐ I found every cited paper myself, by title or DOI.
- ☐ The proof starts from what is known, not from what it wants to prove.
- ☐ I redid the arithmetic with a second method: an estimate, a CAS, or by hand.
- ☐ I plotted the function (Desmos or GeoGebra) and the picture agrees.

> If you cannot tell whether a step is right, that is the step to ask about. Use the prompt *Find the error in this step* from Guide 1, and then check the reply with this list too.